

Report on "S&P500 project"

Intro

- This is a beginner project on finance and visualization. In this project I have chosen S&P500 data from yfinance module to visualize on and take insights from them. For this I have touched some statistics topics like normalization test, moments, financial behavioural aspect and knowledge of market.

Methods

- For this project I have solely used python and its some of libraries like pandas, numpy, yfinance, scipy, matplotlib etc. I have used normalization test as its primary way to read the data and its underlying structure in order to connect it to human behaviour as well. In short, statistics & visualization and behavioural finance are the core of this project.

Results

- The insights got out of these visualizations are useful to study market and its affect on behaviour of human and vice versa. The first visual i.e "Rolling Returns of 12 months" give us the idea of the over all roll (movement) of market on year basis. i.e 252 days. From this figure we can see the data is slightly left-skewed and the average is mostly around \$8 to \$15. It means on an average, we can get profit of \$8 to \$15 but market is more prone to extremity of losses than gains. Its JB value is 5.2058 which signs to reject normality but p-value



Points on lack of confidence to reject the normality

Figure 2: 'Daily returns' is a histogram on daily returns which has a very high kurtosis and JB value with very low p-value which directly suggests that the data is non-normal thus more prone to extreme values.

As the normality can't be rejected at rolling returns, we used Q-Q plot to examine the normality. From Fig 3: "Q-Q plot for rolling return", we can notice that the left tail is more flat and a sudden drop near -2 which indicates that it is more prone to huge sudden loss and right tail is slightly higher which shows spikes, rallies occurrence in market. Where as Fig 4: "Q-Q plot for daily return" shows inverted S-pattern which conveys the returns are leptokurtic i.e. have more of flat tails. Thus, daily returns are also volatile and can be dragged to extremes more often.

Fig 5: "Rolling kurtosis" and Fig 6: "Skewness-Rolling" jointly gives us idea about the past market regimes. Mostly the ~~are~~ events are extremes as kurtosis is rarely normal. Moreover, mean of the kurtosis is negative which says that mostly the market is (platykurtic) normal and steady, but the frequency of excessively high kurtosis over 20-year period is high and frequent. From rolling skewness, the data is saying that, mostly market is negatively skewed thus, loss oriented. If we use rolling kurtosis and skewness side by side then we can figure out about the market regime. For eg: In between 2008 and 2010 there is a huge positive kurtosis and at the same time the skewness is highly negative. Thus, it says in between 2008 and 2010, there was a huge panic or crisis and it has negatively impacted the return severely. Similarly we can connect the idea as kurtosis represents the intensity of fluctuation and skewness represents (gain or loss) ups and down of market.

Fig 7: "zoomed version of daily return" is a simple zoomed picture of daily return which make us show the occurrence of returns beyond $\pm 3\sigma$ and $\pm 5\sigma$ in real world. Normality says there is less than 0.00002% for occurrence of events greater than $\pm 5\sigma$ but in reality, it has occurred.

Similarly Fig 8: "Volatility Clustering" shows that the extreme events has been followed by extremes more and simple are followed by simple for days. This shows us the frequency of crisis and rallies.

- Discussion

"What does these insight says about behavioural patterns of human?"

- The Jarque-Bera value, p-value of rolling returns shows that about a rolling period of 1 year ~~people are~~ returns are fat tails and similarly daily returns says that it is totally fat tail which says that there is tendency of exceeding normal situations by a market and skewness suggest that mostly market is a negative. Thus, combined it says that people are more affected by loss aversion and panic and crash are ~~more~~ occurring than normal. In the Q-Q plot, we can observe that the left tail is more and suddenly dripped out which implies or suggest us that the people do panic selling and market crashes is more often. Similarly the right tail also suggest us that people also buy excessively ~~when~~ in rallies. Thus, it shows the greed and fear in statistical form. Moreover the rolling kurtosis and skewness helps us to know how the market behaves and it says about peoples' action, the spike ~~on~~ between 2008 to 2010 shows the crisis and panic selling and the same kurtosis over 5 years (from 2010 - 2016) shows the herding behaviour of people, all the people are synchronized thus stability of market.



The volatility cluster charts shows the impact of a crisis or panics. When a crisis hits its continued for a long time which shows that the pain or fear of loss lingers more and the change of volatility cluster shows the change of market regime. From 2008 to 2012, the market is very volatile which depicts the lingering fear of people due to 2008 crisis and from 2012-2020, market is stable which shows people is being stable and secure and then the Covid hits and again volatility increases, another regime of market and thus Covid had made people hopeless so ~~and the~~ way reflect on market. Thus, we can conclude that the behaviour of people directly controls the market and its regime. Thus, human behaviour is one of the essential part of market.